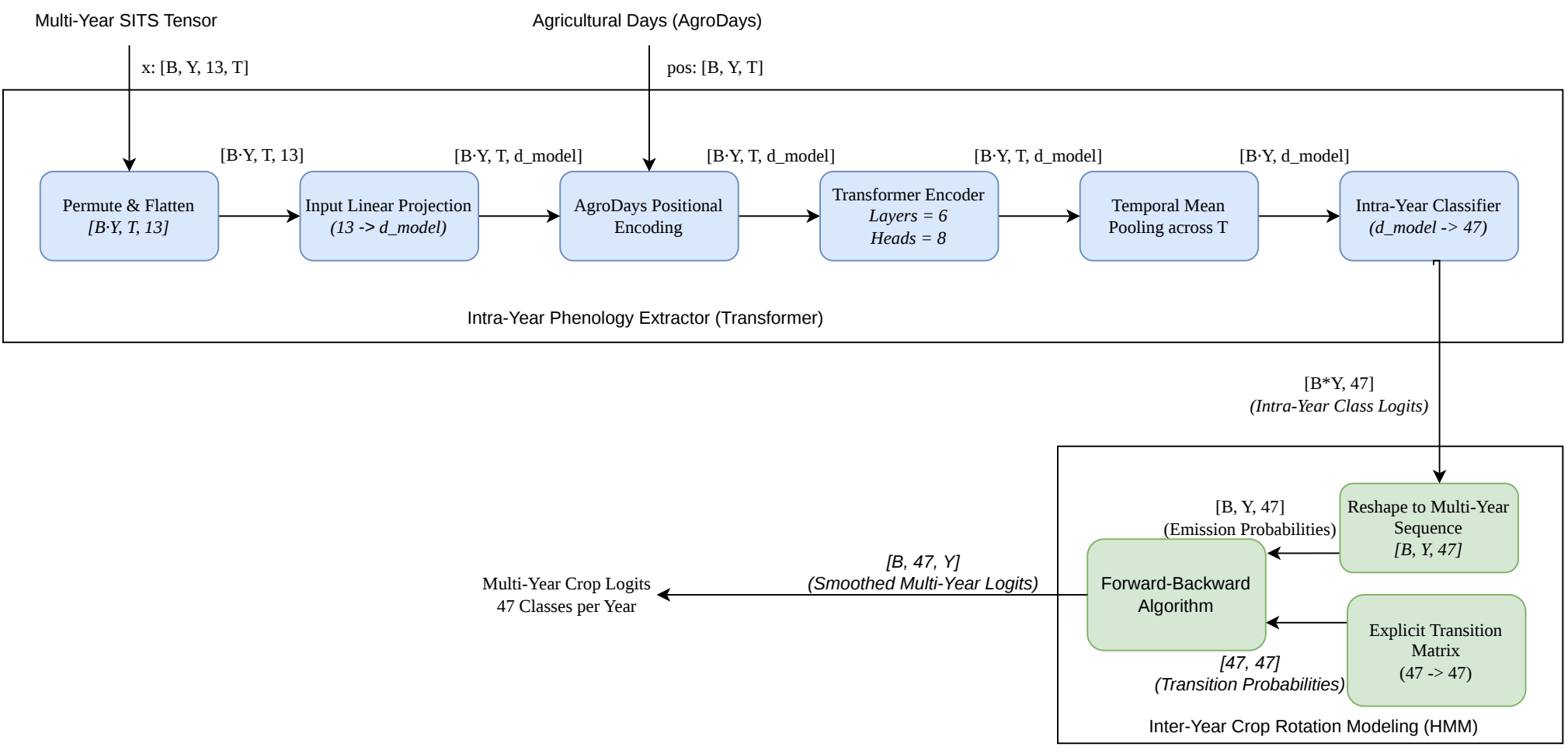

Multi-Year SITS Tensor
x: [B, Y, 13, T]
Agricultural Days (AgroDays)
pos: [B, Y, T]
[B·Y, T, 13]
[B·Y, T, d_model]
[B·Y, T, d_model]
[B·Y, T, d_model]
[B·Y, d_model]
Permute & Flatten
[B·Y, T, 13]
Input Linear Projection
(13 -> d_model)
AgroDays Positional Encoding
Transformer Encoder
Layers = 6
Heads = 8
Temporal Mean Pooling across T
Intra-Year Classifier
(d_model -> 47)
Intra-Year Phenology Extractor (Transformer)
[B*Y, 47]
(Intra-Year Class Logits)
[B, Y, 47]
(Emission Probabilities)
Reshape to Multi-Year Sequence
[B, Y, 47]
[B, 47, Y]
(Smoothed Multi-Year Logits)
Forward-Backward Algorithm
Multi-Year Crop Logits
47 Classes per Year
Explicit Transition Matrix
(47 -> 47)
[47, 47]
(Transition Probabilities)
Inter-Year Crop Rotation Modeling (HMM)